

■ 1. Project Overview

■ Project Title & Objective

Project Title	Olist E-Commerce Business Analysis
Analyst	Prakrit Singh — Data Analyst
Project Period	01 June 2026 – 09 June 2026
Dataset Source	Kaggle — Olist Brazilian E-Commerce Public Dataset
Objective	Clean, transform, and analyze the Olist multi-table e-commerce dataset to answer key business questions and deliver actionable insights via SQL queries and a Power BI dashboard.

■ Business Problem Statement

Olist is a Brazilian e-commerce marketplace that connects small merchants to customers across Brazil. The business generates millions of transactions but lacks consolidated visibility into revenue trends, customer behavior, seller performance, product quality, and logistics efficiency. This project aims to answer critical business questions such as: Which product categories generate the highest revenue? Which states are the most valuable markets? How does delivery time impact customer satisfaction? What payment methods dominate? These insights help Olist and its sellers make smarter, data-driven decisions.

■ Dataset Overview

Table	Description	Raw Rows	Clean Rows	Clean Columns
customers	Customer profile & location	99,441	99,441	4
orders	Order lifecycle & timestamps	99,441	99,441	6
order_items	Line items per order	112,650	112,650	6
payments	Payment method & value	103,886	103,886	4
products	Product catalogue & categories	32,951	32,951	2
reviews	Customer review scores	99,224	99,224	3
sellers	Seller profile & location	3,095	3,095	3

■ ■ Tools Used

Python Python (Pandas, NumPy, Matplotlib, Seaborn)	PostgreSQL Structured Query Language	Power BI Business Intelligence	Olist Data Version: Kaggle Dataset
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■ 2. Dataset & Table Explanation

■ Table 1 — customers

Table Information

Attribute	Detail
Table Name	customers
Purpose	Stores unique customer identities and their geographic location (city, state)
Raw Shape	99,441 rows × 5 columns
Clean Shape	99,441 rows × 4 columns
Important Cols	customer_id (PK), customer_unique_id, customer_city, customer_state

Data Quality Analysis

Quality Check	Result
Total Null Values	0 — No missing data
Duplicate Records	0 — No duplicates
Data Type Issues	None — all types correct
Columns with Nulls	None

Data Cleaning Performed

• **Issue:** Column `customer_zip_code_prefix` was present but not required for analysis. • **Action:** Dropped the column using `customers_df.drop('customer_zip_code_prefix', axis=1)`. • **Why:** ZIP prefix provides no analytical value when city and state columns are already available. • **Result:** Clean table exported as `clean_customer.csv` (99,441 × 4).

■ Table 2 — orders

Table Information

Attribute	Detail
Table Name	orders
Purpose	Tracks the full lifecycle of every order — from purchase to delivery
Raw Shape	99,441 rows × 8 columns
Clean Shape	99,441 rows × 6 columns
Important Cols	order_id (PK), customer_id (FK), order_status, order_purchase_timestamp, order_delivered_customer_date, order_estimated_delivery_date

Data Quality Analysis

Column	Null Count	Reason / Action
order_approved_at	160	Dropped — not needed for analysis

Column	Null Count	Reason / Action
order_delivered_carrier_date	1,783	Dropped — not needed for analysis
order_delivered_customer_date	2,965	Kept — nulls are logically valid (cancelled/undelivered orders)

Data Cleaning Performed

• **Issue 1 — Wrong Data Types:** Five timestamp columns stored as *object (string)* instead of *datetime*. • **Action:** Converted *order_purchase_timestamp*, *order_delivered_customer_date*, and *order_estimated_delivery_date* to *datetime* using *pd.to_datetime()*. • **Issue 2 — Irrelevant Columns:** *order_approved_at* and *order_delivered_carrier_date* had high nulls and were not needed. • **Action:** Dropped both columns to simplify the table. • **Issue 3 — Delivery Nulls (2,965):** Nulls in *order_delivered_customer_date* were intentionally kept because those orders were cancelled, shipped, or unavailable — no actual delivery occurred.

Table 3 — products

Table Information

Attribute	Detail
Table Name	products
Purpose	Product catalogue — maps product IDs to category names
Raw Shape	32,951 rows × 9 columns
Clean Shape	32,951 rows × 2 columns
Important Cols	product_id (PK), product_category_name

Data Quality Analysis

Column	Null Count	Action
product_category_name	610	Filled with 'unknown'
product_name_lenght	610	Dropped
product_description_lenght	610	Dropped
product_photos_qty	610	Dropped
product_weight_g	2	Dropped
product_length_cm	2	Dropped
product_height_cm	2	Dropped
product_width_cm	2	Dropped

Data Cleaning Performed

• **Issue 1 — Irrelevant Columns:** 7 columns (dimensions, weight, photo count, name/description length) were not required for business analysis. • **Action:** All 7 columns dropped — table reduced from 9 to 2 columns. • **Issue 2 — Null Category Names (610 rows):** 610 products (only 0.18% of total) had no category name. • **Action:** Filled nulls with the label 'unknown' using *fillna('unknown')* — preserving those products in analysis rather than discarding them. • **Why:** Dropping 610 rows would remove valid sales data. Labelling as 'unknown' makes it traceable.

Table 4 — payments

Table Information

Attribute	Detail
Table Name	payments
Purpose	Records how customers paid for each order (type, installments, value)
Raw Shape	103,886 rows × 5 columns
Clean Shape	103,886 rows × 4 columns
Important Cols	order_id (FK), payment_type, payment_installments, payment_value

Data Quality Analysis

Quality Check	Result
Total Null Values	0 — No missing data
Duplicate Records	0 — No duplicates
Data Type Issues	None — all types correct
Statistics	All numeric columns are within valid ranges

Data Cleaning Performed

- **Issue:** Column *payment_sequential* was present but had no analytical value.
- **Action:** Dropped *payment_sequential* using *drop()*.
- **Why:** This column tracks installment sequence, which is irrelevant to business analysis.
- **Result:** Clean table saved as *clean_payments.csv* (103,886 × 4).

Table 5 — reviews

Table Information

Attribute	Detail
Table Name	reviews
Purpose	Customer satisfaction ratings and comments for each order
Raw Shape	99,224 rows × 7 columns
Clean Shape	99,224 rows × 3 columns
Important Cols	review_id (PK), order_id (FK), review_score (1–5)

Data Quality Analysis

Column	Null Count	Action
review_comment_title	87,656	Dropped — high nulls, not needed
review_comment_message	58,247	Dropped — high nulls, not needed
review_creation_date	—	Dropped — not required for analysis
review_answer_timestamp	—	Dropped — not required for analysis

Data Cleaning Performed

• **Issue — High Null Text Columns:** `review_comment_title` had 87,656 nulls (88% of rows) and `review_comment_message` had 58,247 nulls (59% of rows). • **Action:** Dropped all 4 non-essential columns. Retained only `review_id`, `order_id`, and `review_score`. • **Why:** Text comment columns could not be reliably used in SQL/Power BI analysis due to massive missing data. The numeric review score is sufficient for performance measurement.

■ Table 6 — sellers

Table Information

Attribute	Detail
Table Name	sellers
Purpose	Seller profile information including city and state
Raw Shape	3,095 rows × 4 columns
Clean Shape	3,095 rows × 3 columns
Important Cols	seller_id (PK), seller_city, seller_state

Data Quality Analysis

Quality Check	Result
Total Null Values	0 — No missing data
Duplicate Records	0 — No duplicates
Data Type Issues	None — all types correct

Data Cleaning Performed

• **Issue:** Column `seller_zip_code_prefix` was present but not required for analysis. • **Action:** Dropped the column using `sellers_df.drop('seller_zip_code_prefix', axis=1)`. • **Why:** ZIP prefix is redundant when city and state are available for geographic analysis.

■ Table 7 — order_items

Table Information

Attribute	Detail
Table Name	order_items
Purpose	Line-item detail for each order — products, sellers, price, freight
Raw Shape	112,650 rows × 7 columns
Clean Shape	112,650 rows × 6 columns
Important Cols	order_id (FK), order_item_id, product_id (FK), seller_id (FK), price, freight_value

Data Quality Analysis

Quality Check	Result
Total Null Values	0 — No missing data

Quality Check	Result
Duplicate Records	0 — No duplicates
Data Type Issues	None — all types correct
Statistics	Price and freight values are within reasonable ranges

Data Cleaning Performed

• **Issue:** Column *shipping_limit_date* was present but irrelevant to business analysis. • **Action:** Dropped *shipping_limit_date* using *drop()*. • **Why:** The analysis focuses on actual delivery performance and revenue — the shipping deadline column is a seller operational field not required for insights. • **Result:** Clean table saved as *clean_order_items.csv* (112,650 × 6).

3. Data Processing & Analysis

Complete Workflow

The project followed a structured data pipeline — from raw CSV files to a production-ready PostgreSQL database and a Power BI dashboard.

Step	Activity	Tools Used	Output
1	Raw data ingestion — 7 CSV files imported from Kaggle	Python / Pandas	7 raw DataFrames
2	Data profiling — shape, dtypes, nulls, duplicates	Pandas (.info, .isna, .duplicated)	Quality report per table
3	Column removal — dropped irrelevant columns across all tables	Pandas .drop()	Leaner DataFrames
4	Data type correction — timestamps converted from object to datetime	pd.to_datetime()	Correct date types
5	Null handling — strategic fill ('unknown') or intentional retain	Pandas .fillna()	Zero analytical nulls
6	Export to clean CSV — all 7 tables saved	DataFrame.to_csv()	7 clean CSV files
7	Database load — clean data pushed to PostgreSQL via SQLAlchemy	SQLAlchemy / psycopg2	olist_db database
8	SQL Analysis — 36 business questions answered	PostgreSQL 15	Business insights
9	Power BI — interactive dashboard created from DB	Power BI Desktop	Published dashboard

Database Setup

Database	olist_db
Engine	PostgreSQL 15 (localhost:5432)
Username	postgres
Connector	SQLAlchemy + psycopg2
Method	DataFrame.to_sql() with if_exists='append' — data appended to pre-created tables
Tables Loaded	customers, sellers, product, orders, payments, reviews, order_items

■ 4. SQL Analysis — Business Questions

A total of **36 SQL queries** were written to answer business questions. These cover basic metrics, join-based analysis, time-series trends, advanced business analysis, and window functions. Key queries with results are documented below.

■ Basic Analysis (Q1–Q10)

Q1. Total Customers

SQL Logic	SELECT COUNT(DISTINCT customer_id) FROM customers
Result	99,441 unique customers
Meaning	The platform has nearly 100K unique customers — a large and diverse customer base.

Q2. Total Orders

SQL Logic	SELECT COUNT(DISTINCT order_id) FROM orders
Result	99,441 total orders
Meaning	Each customer placed at most one order in the dataset, showing a 1:1 order-customer ratio.

Q3. Total Sellers

SQL Logic	SELECT COUNT(DISTINCT seller_id) FROM sellers
Result	3,095 unique sellers
Meaning	A marketplace with ~3,000 active merchants serving nearly 100K customers.

Q4. Total Products

SQL Logic	SELECT COUNT(DISTINCT product_id) FROM product
Result	32,951 unique products
Meaning	Over 32K SKUs across 74 categories — a wide and diverse product catalogue.

Q5. Total Revenue

SQL Logic	SELECT SUM(price) FROM order_items
Result	R\$ 13,591,643.70 total revenue
Meaning	Total platform revenue exceeds R\$13.5 million — strong market performance.

Q6. Average Order Value

SQL Logic	SELECT AVG(order_value) FROM subquery grouping order_id
Result	R\$ 137.75 average order value
Meaning	Customers spend ~R\$138 per order on average — a healthy mid-range basket size.

■ Order Status & Payment Analysis

Q9. Order Status Distribution	
SQL Logic	SELECT order_status, COUNT(*) FROM orders GROUP BY order_status
Result	Delivered: 96,478 Shipped: 1,107 Cancelled: 625 Unavailable: 609 Others: ~622
Meaning	97% of orders were successfully delivered — very high fulfillment rate. Only 0.6% cancelled.

Q10. Most Used Payment Method	
SQL Logic	SELECT payment_type, COUNT(*) FROM payments GROUP BY payment_type ORDER BY COUNT DESC
Result	Credit Card: 76,795 Boleto: 19,784 Voucher: 5,775 Debit Card: 1,529
Meaning	Credit card dominates at 74% of all transactions. Boleto (bank slip) is the #2 method in Brazil.

■ Top Performers — Customers, Sellers, Categories

Q12. Top 10 Customers by Spending	
SQL Logic	JOIN orders → customers → order_items; GROUP BY customer_id; ORDER BY SUM(price+freight) DESC
Result	Top customer (Rio de Janeiro) spent R\$13,664. Top 10 customers spent between R\$4,681–R\$13,664.
Meaning	High-value customers are concentrated in major cities. Loyalty programmes could retain them.

Q13. Top 10 Sellers by Revenue	
SQL Logic	JOIN order_items → sellers; GROUP BY seller_id; ORDER BY SUM(price+freight) DESC
Result	Top seller (Guariba) generated R\$249,640. Top 10 sellers earned between R\$160K–R\$249K.
Meaning	A small group of top sellers drives a large share of revenue — Pareto principle in action.

Q15. Top 10 Product Categories by Revenue	
SQL Logic	JOIN order_items → product; GROUP BY category; ORDER BY SUM(price) DESC
Result	beleza_saude: R\$1.26M relorios_presentes: R\$1.21M cama_mesa_banho: R\$1.04M
Meaning	Health & Beauty and Watches/Gifts are the top revenue categories — premium segment performs well.

Q17. Revenue by Customer State	
SQL Logic	JOIN orders → customers → order_items; GROUP BY state; ORDER BY SUM(price) DESC
Result	SP: R\$5.2M RJ: R\$1.8M MG: R\$1.6M RS: R\$750K PR: R\$683K
Meaning	São Paulo alone contributes 38% of all revenue — a critical market for Olist.

■ Time Series Analysis (Q21–Q25)

Q21. Monthly Order Trend	
SQL Logic	EXTRACT(MONTH FROM order_purchase_timestamp); COUNT(order_id); GROUP BY month
Result	Peak months: Aug (10,843), May (10,573), Jul (10,318). Low months: Sep (4,305), Dec (5,674).
Meaning	Mid-year (May–August) drives the most orders. Seasonal dips occur in Sep–Oct and Dec.

Q22. Monthly Revenue Trend	
SQL Logic	JOIN orders → order_items; EXTRACT month; SUM(price); GROUP BY month
Result	Peak: May R\$1.50M, Aug R\$1.43M. Low: Sep R\$624K, Dec R\$744K.
Meaning	Revenue follows order volume trends — May and August are the best months for revenue.

Q23. Year-over-Year Revenue	
SQL Logic	EXTRACT(YEAR FROM order_purchase_timestamp); SUM(price); GROUP BY year
Result	2016: R\$49K 2017: R\$6.16M 2018: R\$7.39M
Meaning	Revenue grew 20% from 2017 to 2018, showing strong year-over-year platform growth.

Q25. Avg Delivery Time by Month	
SQL Logic	AVG(order_delivered_customer_date - order_purchase_timestamp); GROUP BY month
Result	Fastest: August (8 days), July (9 days). Slowest: February (15 days), December (14 days).
Meaning	Summer months (Jun–Aug) deliver fastest. January–February and Nov–Dec have longer delivery times.

■ Business & Advanced Analysis (Q26–Q36)

Q26. High Revenue, Low Review Score Categories	
SQL Logic	JOIN order_items → product → reviews; HAVING revenue>=800K AND avg_score<=4
Result	cama_mesa_banho (R\$1.24M, 3.90 score) moveis_decoracao (R\$902K, 3.90) informatica_acessorios (R\$1.06M, 3.93)
Meaning	These 3 categories generate high revenue but have customer satisfaction issues — a risk to long-term revenue if not addressed.

Q28. Delivery Time Impact on Review Scores	
SQL Logic	JOIN orders → reviews; GROUP BY review_score; AVG(delivery_date - purchase_date)
Result	5-star: 10 days 4-star: 11 days 3-star: 13 days 2-star: 16 days 1-star: 20 days
Meaning	Clear negative correlation — longer delivery = lower review scores. Improving logistics directly improves customer satisfaction.

Q29. Payment Method Revenue Contribution	
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SQL Logic	JOIN payments → order_items; GROUP BY payment_type; SUM(price+freight)
Result	Credit card: R\$12.72M Boleto: R\$2.84M Voucher: R\$785K Debit: R\$215K
Meaning	Credit card drives 77% of total revenue. Expanding credit options could further increase revenue.

Q33. Running Monthly Revenue (Window Function)

SQL Logic	SUM(revenue) OVER(ORDER BY month_number) — cumulative total
Result	Cumulative revenue grows from R\$1.26M in Jan to ~R\$16.6M by December
Meaning	Steady cumulative growth confirms platform is consistently scaling month over month.

Q35. Month-over-Month Revenue Growth

SQL Logic	LAG(revenue) OVER(ORDER BY month_no) — calculate % change month to month
Result	Biggest growth: Jan→Mar (+24%). Biggest drop: Aug→Sep (-55% — likely data cutoff effect).
Meaning	The Sep–Oct dip may reflect dataset recency cutoff rather than actual business decline.

■ 5. Python Analysis

■ Libraries Used

Library	Version	Purpose
Pandas	Latest	Data loading, cleaning, transformation, export
NumPy	Latest	Numerical operations and array support
Matplotlib	Latest	Base plotting library
Seaborn	Latest	Statistical visualizations
SQLAlchemy	Latest	Database engine connection to PostgreSQL
psycopg2	Latest	PostgreSQL adapter for Python

■ Data Cleaning Process

The cleaning process followed a consistent protocol for each of the 7 tables:

Step 1 — Data Inspection	Used <code>.head()</code> , <code>.info()</code> , <code>.describe()</code> , <code>.isna().sum()</code> , <code>.duplicated().sum()</code> , <code>.shape</code> , <code>.columns</code> for complete profiling of each table.
Step 2 — Column Removal	Dropped columns that had no analytical value: <code>customer_zip_code_prefix</code> , <code>order_approved_at</code> , <code>order_delivered_carrier_date</code> , product dimensions/weight, review text columns, <code>payment_sequential</code> , <code>shipping_limit_date</code> .
Step 3 — Data Type Fix	Converted 3 timestamp columns in <code>orders_df</code> from object to datetime using <code>pd.to_datetime()</code> for time-based analysis.
Step 4 — Null Handling	610 null product category names filled with 'unknow' (only 0.18% of data). Delivery date nulls in orders kept intentionally — logically valid for undelivered orders.
Step 5 — Export	All 7 cleaned DataFrames saved as CSV files using <code>.to_csv(index=False)</code> .
Step 6 — DB Load	Used SQLAlchemy <code>create_engine()</code> and <code>.to_sql()</code> to push all 7 tables to PostgreSQL <code>olist_db</code> for SQL analysis.

■ Key Python Findings

- 99,441 customer records — zero nulls, zero duplicates — highly clean source data.
- Orders table had 5 datetime columns incorrectly typed as strings — fixed before SQL analysis.
- Products table reduced from 9 columns to just 2 (`product_id` + `category_name`) — kept only what matters.
- Reviews table had 88% null rate in comment title and 59% in comment message — text columns unsuitable for analysis.
- The database load step (SQLAlchemy → PostgreSQL) enabled seamless SQL querying and Power BI connectivity.
- Total data volume across all tables: ~47.7 MB compressed / ~550K+ rows across all tables.

■ 6. Power BI Dashboard Analysis

A Power BI Desktop dashboard (.pbix) was created by connecting directly to the PostgreSQL olist_db database. The dashboard provides interactive visual analysis of the cleaned Olist data.

■ Dashboard Objective

- Provide a single interactive view of Olist's business performance.
- Enable stakeholders to explore revenue, orders, sellers, and customer satisfaction by category, state, and time.
- Support quick identification of top-performing and underperforming areas.
- Allow dynamic filtering by product category, customer state, payment type, and time period.

■ Key KPIs Tracked

13.59M Total Revenue (R\$)	99,441 Total Orders	99,441 Total Customers	3,095 Total Sellers
137.75 Avg Order Value (R\$)	4.07 / 5 Avg Review Score	32,951 Total Products	97% Delivery Rate

■ Charts & Visuals Explanation

Visual	What It Shows	Key Insight
Revenue by Category (Bar Chart)	Total revenue per product category, sorted descending	beleza_saude leads at R\$1.26M — health & beauty is the top earner
Monthly Revenue Trend (Line Chart)	Revenue per month across all years combined	May and August peak months; September shows sharp dip
Order Status Pie Chart	Distribution of all order statuses	97% delivered — strong operational fulfillment
Payment Method Bar Chart	Count and revenue by payment type	Credit card: 74% of all transactions
Revenue by State (Map / Bar)	Total customer revenue per Brazilian state	SP dominates with 38% of total revenue
Review Score Distribution	Count of reviews per score (1–5)	Majority of reviews are 5-star — generally satisfied customers
Delivery Time vs Review Score	Average delivery days grouped by review score	Faster delivery (10 days) = 5 stars; slow delivery (20 days) = 1 star
Top Sellers by Revenue (Table/Bar)	Top 10 sellers with their revenue and city	Top seller (Guariba) earned R\$249K — 80x more than bottom sellers
Year-over-Year Revenue (Column Chart)	Annual revenue: 2016, 2017, 2018	Revenue grew from R\$6.16M (2017) to R\$7.39M (2018) — +20% growth
Categories with High Revenue / Low Score	Scatter plot: revenue vs avg review score	cama_mesa_banho and informatica_acessorios need quality improvement

■ Filters & Slicers

• **Product Category Slicer:** Filter all visuals by one or more product categories. • **Customer State Slicer:** Focus on specific Brazilian states (e.g., SP, RJ, MG). • **Payment Type Filter:** Isolate credit card, boleto, voucher, or debit transactions. • **Order Status Filter:** Filter by delivered, cancelled, shipped, etc. • **Year / Month Slicer:** View trends for 2016, 2017, or 2018 independently.

■ 7. Key Insights

■ Revenue & Business Performance

- Total platform revenue of R\$13.59M across 99,441 orders — average order value of R\$137.75.
- Revenue grew 20% from 2017 (R\$6.16M) to 2018 (R\$7.39M), confirming strong business growth.
- São Paulo state contributes 38% of total customer revenue — it is the single most important market.
- Credit card payment drives 77% of revenue (R\$12.72M) — dominance of this payment method is clear.
- Top 3 revenue categories: beleza_saude (R\$1.26M), relorios_presentes (R\$1.21M), cama_mesa_banho (R\$1.04M).

■ Customer Satisfaction & Delivery

- Average review score across all products is 4.07/5 — generally positive customer experience.
- Strong delivery-review correlation: 5-star orders delivered in 10 days vs 1-star in 20 days.
- August has the fastest average delivery (8 days), while February is slowest (15 days).
- 97% order delivery success rate — only 0.6% of orders cancelled.
- Categories cds_dvds_musicais (4.64) and fashion_roupa_infanto_juvenil (4.50) have best review scores.
- seguros_e_servicos (2.50) and pc_gamer (3.33) have the lowest average review scores.

■ Seller Performance

- Top seller (Guariba) generated R\$249,640 — 80x more than the bottom seller (R\$12).
- São Paulo state generates the most seller revenue (R\$8.75M) — a major hub for merchants.
- 3,095 sellers serve 99,441 customers — average of ~32 customers per seller.
- Large performance gap between top and bottom sellers signals need for seller support programs.

■ Risk Areas & Weak Points

- 3 high-revenue categories (cama_mesa_banho, moveis_decoracao, informatica_acessorios) have review scores below 4.0 — quality issues in top-revenue segments are a business risk.
- Delivery times in Q1 (Jan–Feb) and Q4 (Nov–Dec) are significantly longer — festive season logistics need improvement.
- September shows a sudden 55% revenue drop — could be a dataset completeness issue or seasonal effect.
- Bottom 10 sellers earn under R\$25 — low-performing sellers drag marketplace average down.
- Voucher and debit card transactions are minimal (R\$785K and R\$215K) — underutilized payment channels.

■ Seasonal & Time Patterns

- Peak order months: August (10,843), May (10,573), July (10,318).
- Lowest order months: September (4,305), December (5,674) — despite Dec being festive season.
- Weekday analysis: Monday sees the most orders (16,196) — customers shop early in the week.
- Saturday and Sunday have the lowest order volumes — weekend engagement is lower.

■ 8. Business Recommendations

■ What Is Working Well

- 97% order delivery rate — the fulfillment engine is highly reliable. Maintain logistics partnerships.
- Health & Beauty category (beleza_saude) is the top revenue earner — invest more in this segment.
- Credit card dominance (74%) means payment infrastructure is well-utilized.
- Year-over-year 20% revenue growth is strong — the platform's growth trajectory is healthy.
- São Paulo market is extremely well-served — continue to strengthen presence there.

■ What Needs Improvement

- Delivery times in January–February and November–December are 14–15 days — target under 10 days.
- cama_mesa_banho and informatica_acessorios generate high revenue but have 3.90–3.93 scores — product quality/seller standards must improve in these categories.
- Bottom 10% of sellers contribute minimal revenue — provide training, analytics, or incentives.
- September–October revenue dip needs investigation — seasonal marketing campaigns should fill this gap.
- Debit card payments are severely underutilized (R\$215K) — explore why customers avoid this option.

■ Recommended Actions

Priority	Recommendation	Expected Impact
■ HIGH	Launch delivery improvement initiative in Q1 and Q4 — reduce avg delivery to under 12 days	Increase review scores by 0.3–0.5 points
■ HIGH	Introduce quality standards for cama_mesa_banho and informatica_acessorios sellers	Improve review scores → protect R\$2M+ in revenue
■ MED	Run targeted marketing campaigns in September–October to fill seasonal revenue gap	Potential R\$300–400K additional revenue
■ MED	Create seller performance tiers — reward top sellers, support bottom sellers with analytics	Improve marketplace average revenue
■ MED	Offer exclusive deals or loyalty rewards for Monday shoppers to maintain peak-day engagement	Sustain high weekday order volume
■ LOW	Expand boleto and debit card promotions to capture more payment variety	Increase transaction volume by 5–10%
■ LOW	Open marketing campaigns targeting RJ, MG, RS — high-potential states after SP	Diversify revenue concentration away from SP

■ Growth Opportunities

- Expand into underserved states — states like AC, AP, RR generate under R\$20K each. Targeted campaigns could unlock significant new revenue.
- Saturday/Sunday orders are lowest — weekend promotions, flash sales, or weekend-exclusive deals could boost weekly revenue by 10–15%.
- relgios_presentes (Watches & Gifts) is the #2 revenue category with a 4.02 score — a prime candidate for premium category promotion.
- The 20% YoY growth trend suggests strong platform-market fit — now is the time to scale seller onboarding and marketing investment.

■ 9. Project Summary

■ What Problem Was Solved

Olist's multi-table e-commerce data was raw, inconsistent, and scattered across 7 CSV files with data type issues, unnecessary columns, and missing values. This project consolidated, cleaned, and transformed all data into a production-ready format for business analysis.

■ What Analysis Was Performed

Tool	Analysis Done	Volume
Python / Pandas	Data cleaning, profiling, transformation, database loading	7 tables cleaned
PostgreSQL	36 business SQL queries — basic, join, time-series, window functions	36 queries
Power BI	Interactive dashboard with KPIs, charts, slicers, trend analysis	1 dashboard (.pbix)

■ Major Findings

- Total revenue: R\$13.59M across 99,441 orders — average basket size R\$137.75.
- Revenue grew 20% year-over-year from 2017 to 2018.
- São Paulo generates 38% of all customer revenue — the dominant market.
- Delivery speed is the #1 driver of customer satisfaction (10 days = 5 stars, 20 days = 1 star).
- Health & Beauty is the top revenue category (R\$1.26M); Watches & Gifts is #2 (R\$1.21M).
- Credit card drives 77% of total revenue.
- 97% of all orders are successfully delivered — excellent fulfillment rate.
- 3 high-revenue categories have below-4.0 review scores — quality gap risk.
- Top seller earns R\$249K vs bottom seller's R\$12 — massive performance disparity.

■ Business Impact & Final Outcome

This project delivers a complete, end-to-end analytical solution for Olist's e-commerce operations. Stakeholders now have a single source of truth — a clean database + interactive Power BI dashboard — that surfaces revenue patterns, seller performance, customer satisfaction drivers, and seasonal trends. The actionable recommendations directly address quality issues in high-revenue segments, delivery time improvement opportunities, and untapped market segments in underserved Brazilian states.

■ Interview Summary (Tell Me About Your Project)

- I analyzed the Olist Brazilian e-commerce dataset — a real-world, multi-table dataset from Kaggle covering 99,441 orders, 3,095 sellers, 32,951 products, and 7 interlinked tables.
- I started with Python and Pandas to clean, profile, and transform all 7 tables — fixing data types, removing irrelevant columns, and handling null values logically.
- I then loaded the clean data into a PostgreSQL database using SQLAlchemy and wrote 36 SQL queries spanning basic metrics, multi-table joins, time-series trends, and advanced window functions.
- Finally, I built a Power BI dashboard connected to the PostgreSQL database showing KPIs, revenue trends, seller performance, category analysis, and delivery impact on review scores.
- Key finding: Delivery time is the biggest driver of customer satisfaction — and 3 of the highest-revenue categories have below-average review scores, representing a major business risk and improvement opportunity.
- This project demonstrates my ability to work across the full data analyst stack: data cleaning in Python, business analysis in SQL, and storytelling with Power BI.